TECHNICAL REPORT v0.1 · PRELIMINARY · NOT PEER-REVIEWED

Pennywise: Learning to Route LLM Calls with an Open Decision Model

Policy-gradient training of Laya as a cost-aware router

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Built 2026-10-27 from commit `eb720a3-dirty` · seed 0 · all numbers generated from evaluation outputs

Abstract. Agents send every request to a frontier model, including the many bounded ones a far cheaper model would answer correctly. We treat model-tier selection as a contextual bandit and use an open-weight, non-autoregressive decision model, Laya [1], as the policy: its distribution over {small, medium, large} *is* $\pi(a|s)$, available from one forward pass. After a supervised warm start on hindsight-optimal labels, we train one policy per cost weight λ with REINFORCE on $r = \text{correct}(a) - \lambda \cdot c(a)/c_{\text{ref}}$, observing only the sampled arm. We build an outcome table of 948 automatically graded prompts answered by three open-weight tiers (2844 calls; tier accuracies 74.5 / 83.8 / 84.0%, any-tier ceiling 91.6%). On a held-out test set of 189 prompts, the validation-selected `auto` policy ($\lambda = 0.5$) reaches **81.5%** accuracy (95% CI 76.2–86.8) at **2.0%** of the cost of always calling the largest model (CI 1.8–2.3), versus 84.1% for always-large; the paired accuracy difference is -2.6 points (CI -7.9 to +2.6). The trained policies mostly learn to default to the medium tier, close to an always-medium baseline (difference -1.1 points, CI -2.6 to +0.0); we report confidence intervals throughout and compare against strong cheap baselines, including where they win. On an agent coding benchmark against Alan Code's Auto Router, Pennywise with a tool-call cascade merges 10/10 unit-tested tasks, the same as Auto Router (10/10), and finishes them 3.4× faster (11.0 vs. 36.9 minutes).

1 Introduction

Coding agents and LLM applications route nearly every call, from classification and summaries to multi-step reasoning, to one expensive model. Teams that measure AI spend in *dollars per merged pull request* pay frontier prices for requests a small open model would answer correctly. On our outcome table the largest tier costs on average 101× more per call than the small tier and 43× more than the medium tier, yet the medium tier's accuracy (83.8%) is within a point of the large tier's (84.0%). LLM routing addresses this gap by choosing a model per request [2], [3], [4].

A recent class of *decision models* returns typed answers with probabilities instead of generating text: the closed Jev [5] and the open-weight Laya [1]. Such a model's output over a small, typed option set is exactly a stochastic policy, so it can be trained directly with policy gradients, without token-level RL.

Contributions (scoped to what we measured):

1. A router that uses Laya's option distribution as the policy $\pi_\theta(a|s)$ over three model tiers, trained with a supervised warm start followed by REINFORCE on a correctness-minus-cost reward, yielding a family of policies indexed by λ (Section 4).
2. An outcome table of 948 automatically graded prompts (math, science QA, code, and an agent-style slice of bug-fix and CI-triage tasks built by bug injection), each answered by all three tiers (Section 5), with bootstrap confidence intervals and cheap embedding baselines, including per-tier correctness predictors swept over the same λ grid (Section 6).
3. An open implementation: an OpenAI-compatible proxy with sticky per-session routing and tool-call passthrough, so agent CLIs can use the router by changing one base URL (Section 9).
4. A tool-call cascade for agent sessions that validates every cheap-tier step and escalates on evidence of failure, and a head-to-head agent benchmark against Alan Code's Auto Router (Section 8).

2 Related work

Routing and cascades. RouteLLM trains routers from preference data to choose between a strong and a weak model [2]. FrugalGPT learns cascades over multiple LLMs to reduce cost [3]. AutoMix combines small-model self-verification with a POMDP-based router that decides when to escalate [4]. RouterBench provides a benchmark and a large dataset of inference outcomes for evaluating routers [6]; we have not evaluated on it yet.

Decision models. Jev is a closed decision model returning typed choices, scores and yes/no answers with probabilities [5]. Laya is an open-weight model of the same category: a ModernBERT-large encoder [7] with a decision head that scores each option at its own marker in one pass [1]. REFLEX uses Jev as a fast typed decision layer inside an agent and escalates to a strong LLM when confidence is low [8]. We instead *train* the decision model as a routing policy with RL on observed outcomes, using open weights end to end.

3 Problem formulation

Let s be a request and $\mathcal{A} = \{\text{small}, \text{medium}, \text{large}\}$ the model tiers, ordered by price. Calling tier a on s yields a graded outcome $y(s,a) \in \{0,1\}$ and a cost $c(s,a)$ in USD, computed from the call's measured token counts at the tier's public list price. A router is a policy $\pi(a|s)$. For a cost weight $\lambda \geq 0$ we maximise the expected reward

$$J_\lambda(\pi) = \mathbb{E}_{s\sim\mathcal{D}}\mathbb{E}_{a\sim\pi(\cdot|s)}\left[y(s,a) - \lambda \cdot \frac{c(s,a)}{c_{\text{ref}}}\right], \qquad (1)$$

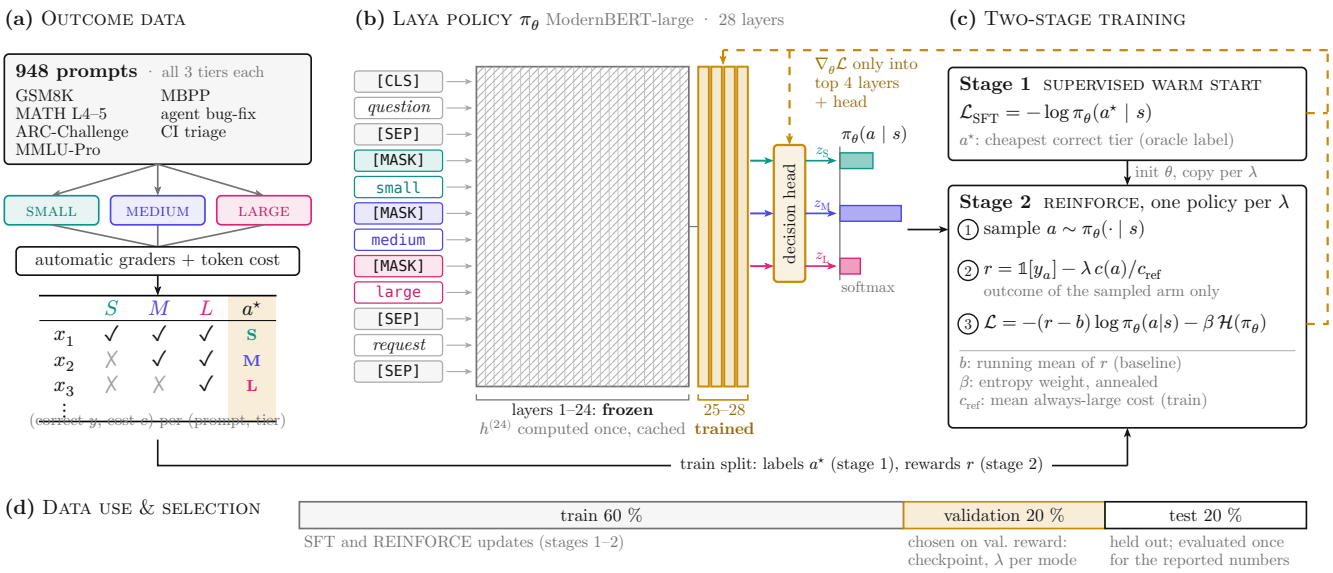


Figure 1. Training pipeline. (a) Every prompt is answered by all three tiers and graded automatically. (b) Laya scores the three tier options at their `[MASK]` markers in one non-autoregressive pass; the bottom 24 encoder layers are frozen and their outputs cached, the top 4 layers and the decision head are trained. (c) A supervised warm start on oracle labels is followed by REINFORCE per λ, using only the sampled arm's reward. (d) Checkpoints and proxy modes are selected on validation; the test split is used once.

where $c_{\text{ref}} = 0.003138$ USD is the mean cost of always-large on the training split, which makes λ unitless: $\lambda = 1$ trades one correct answer for the cost of one average large-tier call. The *oracle* label $a^*(s)$ is the cheapest tier with $y(s, a) = 1$ (or `large` if none); it maximises Equation 1 for any $\lambda > 0$ in hindsight but is not observable at decision time.

This is a contextual bandit: the router sees s, picks one arm, and in deployment observes only that arm's outcome. Our outcome table contains all three outcomes for every prompt, which allows exact offline evaluation; training nevertheless uses only the sampled arm, as an online system would.

4 Method

4.1 Decision-model output as a policy

Laya encodes `[CLS] choice question: <instructions> [SEP] [MASK] opt₀ [MASK] opt₁ [MASK] opt₂ [SEP] <request> [SEP]` and reads one logit z_a per option at its `[MASK]` position [1]. With the three tier options, each given a short description, the softmax

$$\pi_\theta(a|s) = \exp(z_a(s)) / \sum_{a'} \exp(z_{a'}(s)) \qquad (2)$$

defines the policy. One forward pass yields the full distribution, so $\log \pi_\theta(a|s)$ is available for every action at the cost of a single encoder call (Figure 1 b).

4.2 Supervised warm start

We first fine-tune on the oracle label with cross-entropy over the three options, for up to 10 epochs with early stopping on validation loss. To fit a 4 GB GPU we train only the top 4 encoder layers plus the decision head. Because the lower layers are frozen, their outputs are computed once per prompt and cached; training then runs only the trainable layers, which is identical to the full forward pass up to bf16 rounding.

Input: warm-start θ_0, train/val prompts, outcome table (y, c), λ
$b \leftarrow 0$, $\theta \leftarrow \theta_0$
for $t = 1 \ldots T$ **do**
  sample batch $\{s_i\}$; $a_i \sim \pi_\theta(\cdot\,|s_i)$ — *one forward pass*
  $r_i \leftarrow y(s_i, a_i) - \lambda c(s_i, a_i)/c_{\text{ref}}$ — *sampled arm only*
  $\theta \leftarrow \theta - \eta \nabla_\theta \frac{1}{n} \sum_i \mathcal{L}_i$ (Equation 3) — *top 4 layers + head*
  $b \leftarrow 0.95b + 0.05\bar{r}$; anneal β
  every few steps: keep θ if val reward improves
return best θ

Algorithm 1. REINFORCE routing policy for one cost weight λ.

4.3 REINFORCE with baseline and entropy

Starting from the warm start, for each $\lambda \in \{0, 0.1, 0.25, 0.5, 1, 2\}$ we train a separate policy for 800 steps with batches of 16 training prompts, minimising

$$\mathcal{L}(\theta) = -(r - b) \log \pi_\theta(a|s) - \beta H(\pi_\theta(\cdot\,|s)), \qquad (3)$$

with $a \sim \pi_\theta(\cdot\,|s)$ sampled from the current policy, r the sampled arm's reward from Equation 1, b an exponential running mean of the batch reward [9], and β annealed linearly from 0.05 to 0.005. The checkpoint with the highest validation reward is kept (Algorithm 1).

Proxy modes. Three modes map to trained λ values by fixed rules applied to validation results: `auto` is the cheapest RL policy whose accuracy is not significantly below always-large; `auto-quality` the most accurate; `auto-cheap` the cheapest at least as accurate as always-medium. Here `auto` $= 0.5$, `auto-quality` $= 0$, `auto-cheap` $= 0.5$.

4.4 Agent sessions: a tool-call cascade

The router is trained on single-turn prompts, but agent CLIs run multi-turn tool loops. Routing once per session, an agent can land on a tier whose *tool calling* is unreliable even when its answers are good. For routed requests that carry `tools`, the proxy therefore holds back each cheap-tier turn and checks it before the agent sees it: every tool call must name a tool from the request, its arguments must be valid JSON without garbled characters or abbreviated file paths, the turn must not be empty, the upstream must not report an error, and the turn must finish within a time budget (30 s). If any check fails, that turn is replayed on

Table 1. Outcome table by source. Accuracy is over all splits; the oracle label is the cheapest tier that answered correctly ("none": no tier did). Output-token medians show the reasoning length each tier spends.

			accuracy %			oracle label %				median out-tokens		
Source	**n**	**tr/va/te**	S	M	L	S	M	L	none	S	M	L
GSM8K	100	60/20/20	92.0	94.0	94.0	92.0	3.0	3.0	2.0	260	259	342
ARC-C	100	60/20/20	94.0	99.0	97.0	94.0	5.0	0.0	1.0	83	155	280
MBPP	200	120/40/40	76.5	90.0	88.0	76.5	15.0	3.5	5.0	153	396	609
MATH L4–5	150	90/30/30	60.0	78.7	76.7	60.0	23.3	2.7	14.0	1086	1084	1385
MMLU-Pro	200	120/40/40	69.5	71.0	81.0	69.5	11.0	8.0	11.5	310	408	863
Agent bug-fix	98	59/20/19	77.6	85.7	73.5	77.6	13.3	1.0	8.2	107	627	1645
Agent triage	100	60/20/20	62.0	77.0	80.0	62.0	18.0	5.0	15.0	78	167	690
All	948	569/190/189	74.5	83.8	84.0	74.5	13.3	3.8	8.4	208	379	696

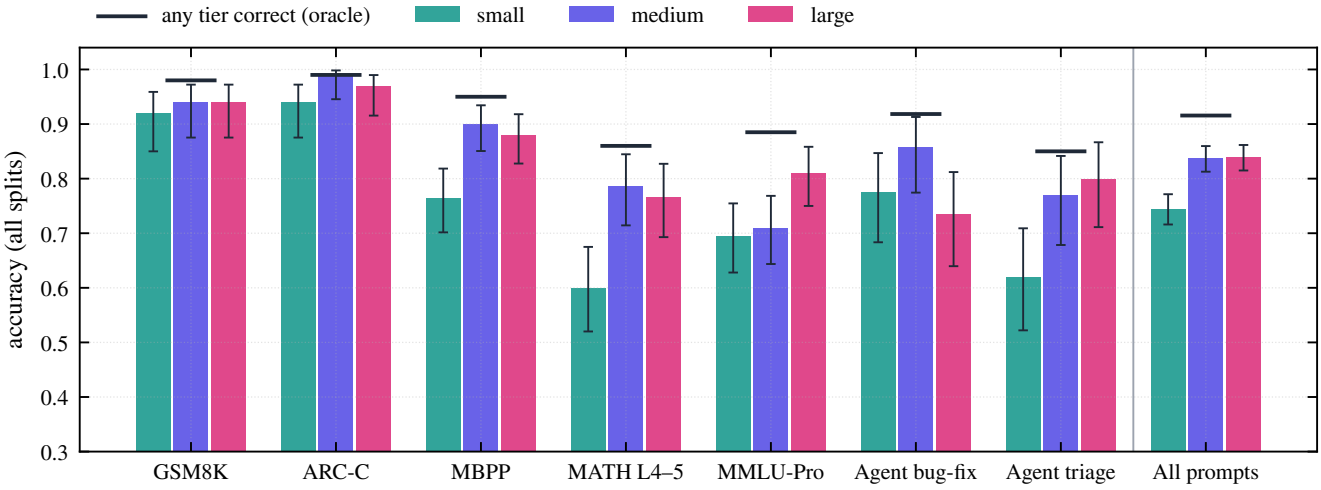


Figure 2. Accuracy per source and tier (all 948 prompts, Wilson 95% intervals). Black ticks: fraction of prompts answered correctly by at least one tier, the ceiling for any router. The medium tier matches or beats the large tier on most sources; the large tier's advantage concentrates on MMLU-Pro and agent triage.

the top tier and the session stays there. This is cheap-first, escalate-on-evidence cascading in the spirit of FrugalGPT [3], applied per step rather than per query.

5 Data and setup

Prompts. 948 prompts with automatically checkable answers, mixing easy and hard sources because a pilot showed current open models saturate the easy ones: GSM8K [10] (final number, exact match), MATH-500 levels 4–5 (the subset [11] of MATH [12]; last `\boxed{}` answer after LaTeX normalisation), ARC-Challenge [13] and MMLU-Pro [14] (option letter), MBPP [15] (unit tests in a sandboxed subprocess, 5 s timeout), and an agent-style slice built from disjoint, longer MBPP functions: *bug-fix* prompts (one or two injected bugs; graded by the original tests) and *CI-triage* prompts (classify the injected bug type from the code and real test output). Splits are stratified by source, 569 / 190 / 189 (train / val / test).

Model tiers. All three tiers are open-weight models with tool calling, served through OpenAI-compatible endpoints (prices per million tokens, input / output, USD):

Tier	Model	In	Out
small	nvidia/nemotron-3.5-bolt-e4b	0.04	0.05
medium	openai/gpt-oss-20b	0.018	0.09
large	nvidia/nemotron-3-ultra-550b-a55b	0.6	2.4

All 2844 calls ran on a free API (actual spend $0), so every cost is the public list price of the same model on a paid provider; the whole table would have cost $3.16 (0.54 M input, 2.45 M output tokens). The small tier has no paid listing; its price is a *proxy*, the list price of a model of the same ~4B size class.

The tiers are complementary rather than nested (Figure 3): 66.2% of prompts are solved by all three tiers and 8.4% by none, while 36 prompts are solved *only* by the large tier and 57 are solved by the small tier but missed by the large one. Routing therefore has headroom (any-tier ceiling 91.6% vs. 84.0% for always-large) but cannot be reduced to a difficulty threshold.

Baselines. Always-small/medium/large; uniform random; the oracle (hindsight, not achievable); zero-shot Laya; logistic regression and k-NN predicting the oracle tier from frozen embeddings (base Laya, mean-pooled;

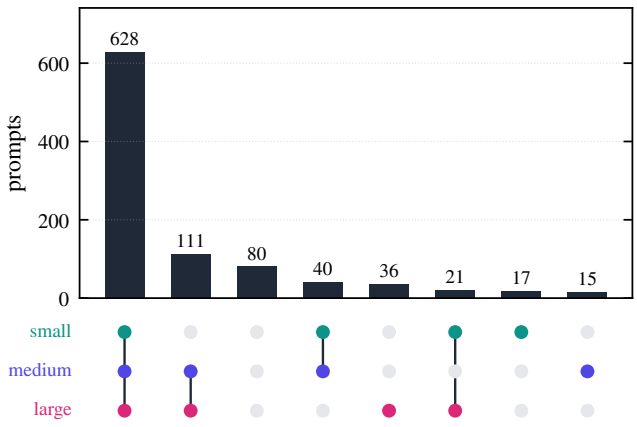


Figure 3. Correctness patterns over the three tiers (dots: tier correct). 8 of 8 possible patterns occur.

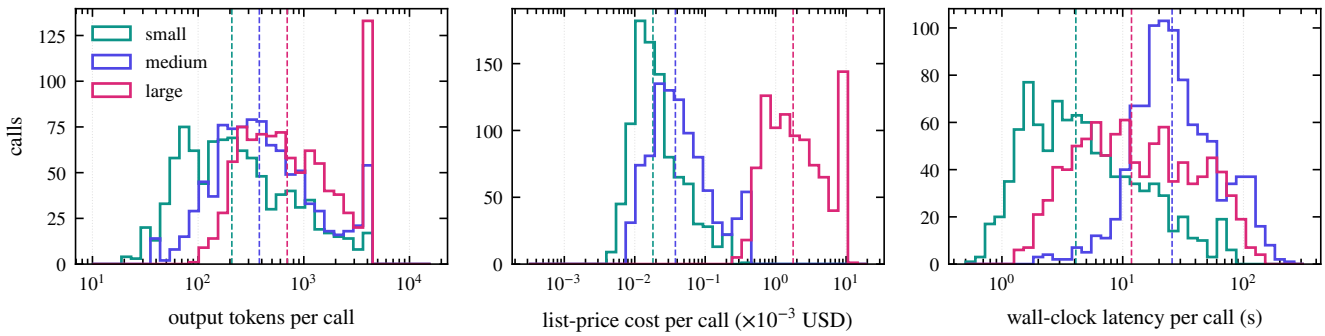


Figure 5. Per-call distributions by tier (dashed: medians). Output length dominates cost; the spike at the right of the large tier's token histogram is the generation cap. Median wall-clock latency on the free API: small 4.1 s, medium 25.6 s, large 11.8 s (shared, rate-limited endpoints; not a serving benchmark).

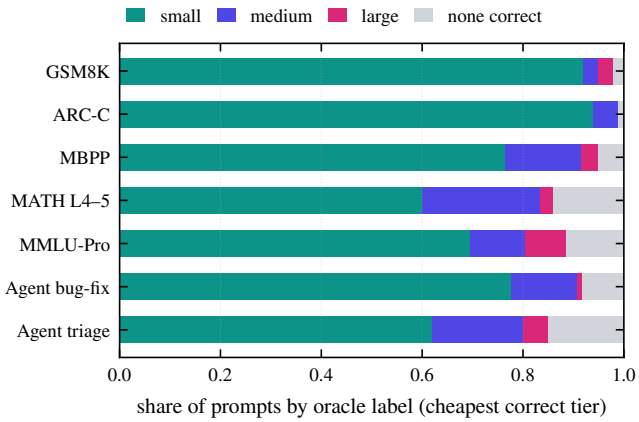


Figure 4. Oracle-label mix per source. Most prompts are solvable by the small tier; the oracle needs the large tier rarely.

and all-MiniLM-L6-v2 [16]), with C and k tuned on validation; and per-tier correctness predictors, one logistic regression per tier estimating $P(\text{correct}|s,a)$, acting by $\arg\max_a P(\text{correct}|s,a) - \lambda \bar{c}_a / c_{\text{ref}}$ over the same λ grid. Jev: not run (no JEV_API_KEY).

Metrics. Test accuracy, mean cost per request, cost relative to always-large, and router latency, each with 95% percentile bootstrap intervals (1000 resamples); comparisons against always-large use paired bootstrap differences.

6 Results

The validation-selected `auto` policy ($\lambda = 0.5$) differs from always-large in accuracy by -2.6 points (paired 95% CI -7.9 to +2.6) while costing 2.0% as much (Table 3). Zero-shot Laya reaches 77.2% at 20.1% cost, and the supervised warm start alone collapses to the majority oracle label (small).

What RL actually learned. Every policy with $\lambda > 0$ routes most requests to the medium tier (96.5% on average, Figure 7), the tier whose accuracy is within a point of the large tier's at a small fraction of its price (Figure 2). The learned policy is therefore close to the trivial always-medium baseline: `auto` differs from it by -1.1 points (paired 95% CI -2.6 to +0.0). RL discovered the right default tier from bandit feedback alone, but on this test set it does not reliably find the prompts that need the large tier: its largest per-source loss against always-large is on MMLU-Pro (-20.0 points, n = 40 Figure 8). The most accurate RL policy on the test split is $\lambda = 0.1$ (83.1% at 1.9% cost); it was not selected because selection uses validation only.

Figure 6 places the RL family against the per-tier correctness predictors, the strongest cheap baseline. At matched λ the two are within confidence intervals of each other; where the baselines match or beat the RL policies, the Laya router's remaining advantages are architectural (a single model scoring options directly, trainable from bandit feedback) rather than accuracy.

Table 2. Training configuration.

Encoder	ModernBERT-large, 28 layers, 421M params (Laya)
Trainable	top 4 layers + decision head
SFT	lr 2e-05, batch 8×2, ≤10 epochs, patience 3
RL	REINFORCE, 800 steps, batch 16, lr 5e-06
Baseline b	running mean, momentum 0.95
Entropy β	0.05 → 0.005 (linear)
λ grid	0, 0.1, 0.25, 0.5, 1, 2
Selection	best val reward (checked every 40 steps)
Bootstrap	1000 resamples, seed 0
Hardware	1× RTX 3050 Laptop (4 GB), bf16

7 Analysis

What the router learns. Figure 10 compares the `auto` policy's actions with the oracle label on the test split; it agrees with the oracle on 14.3% of prompts. Off-diagonal mass above the diagonal is money spent on a tier more expensive than necessary; mass below it is accuracy risked on a tier cheaper than the cheapest correct one.

Confidence. Figure 11 relates the policy's probability of its chosen tier to whether that tier answered correctly (point-biserial $r = 0.28$). The policy's probabilities are trained for reward, not calibrated for correctness; we report them as-is and do not use them for escalation.

8 Agent coding benchmark vs. Alan's Auto Router

Alan Code, the coding agent built into the Alan desktop app, offers an **Auto Router** that "lets Alan choose the model". We compared it with Pennywise on 10 agent coding tasks we wrote for this purpose: 3 easy, 3 medium and 4 hard (e.g. an O(1) LRU cache without library shortcuts, a recursive-descent expression parser without `eval`, a broken Dijkstra, RFC 4180 CSV quoting). Each task is a small repository with a bug or a stub and a pytest suite; a reference solution passes and the starting code fails every suite. A task counts as **merged** when its tests pass and the test file is unchanged. Both systems received the same one-line instruction per task, no hints, one attempt. Pennywise ran in OpenCode, the open coding-agent CLI that Alan

Table 3. Test-set results (n = 189). Brackets: 95% bootstrap intervals. Per-tier LR rows are shown for a subset of λ. Router latency: median single-request time on the build machine's GPU. Pareto: not dominated in (accuracy, cost) by another non-oracle policy.

Policy	Accuracy % [95% CI]	Cost % of always-large [95% CI]	Router ms	Pareto
always-small	73.5 [67.2, 79.9]	1.0 [0.8, 1.1]	0.0	✓
always-medium	82.5 [77.2, 87.8]	2.1 [1.8, 2.3]	0.0	
always-large	84.1 [78.8, 88.9]	100.0 [100.0, 100.0]	0.0	
random	79.9 [74.1, 85.2]	41.4 [32.4, 51.5]	0.0	
oracle	93.7 [90.5, 96.8]	26.0 [17.4, 35.1]		
zero-shot Laya	77.2 [71.4, 83.1]	20.1 [12.4, 28.7]	43.9	
LR (Laya-emb)	74.1 [68.3, 79.9]	6.8 [2.6, 12.4]	30.9	
kNN (Laya-emb)	73.5 [67.2, 79.9]	1.1 [0.9, 1.5]	30.9	
per-tier LR (Laya-emb) (λ=0.1)	82.5 [76.7, 87.8]	8.5 [4.2, 13.4]	30.9	
per-tier LR (Laya-emb) (λ=0.25)	81.5 [75.7, 86.8]	2.8 [1.7, 4.5]	30.9	
per-tier LR (Laya-emb) (λ=0.5)	81.5 [75.7, 86.8]	1.8 [1.5, 2.0]	30.9	✓
per-tier LR (Laya-emb) (λ=1)	81.0 [75.1, 86.2]	1.8 [1.5, 2.0]	30.9	✓
LR (MiniLM-emb)	75.1 [68.8, 81.0]	4.8 [1.5, 9.0]	6.4	
kNN (MiniLM-emb)	73.5 [67.2, 79.9]	1.1 [0.9, 1.3]	6.4	
per-tier LR (MiniLM-emb) (λ=0.1)	83.1 [77.8, 88.4]	7.6 [3.9, 12.3]	6.4	
per-tier LR (MiniLM-emb) (λ=0.25)	82.0 [76.7, 87.3]	3.4 [2.0, 5.1]	6.4	
per-tier LR (MiniLM-emb) (λ=0.5)	81.0 [75.1, 86.2]	1.9 [1.6, 2.1]	6.4	
per-tier LR (MiniLM-emb) (λ=1)	81.0 [75.1, 86.2]	1.8 [1.6, 2.1]	6.4	
SFT Laya	73.5 [67.2, 79.9]	1.0 [0.8, 1.1]	43.9	✓
RL Laya (λ=0)	83.1 [77.2, 88.4]	86.2 [78.8, 92.4]	43.9	
RL Laya (λ=0.1)	83.1 [77.8, 88.4]	1.9 [1.7, 2.2]	43.9	✓
RL Laya (λ=0.25)	82.0 [76.7, 87.3]	2.0 [1.8, 2.3]	43.9	
RL Laya (λ=0.5)	81.5 [76.2, 86.8]	2.0 [1.8, 2.3]	43.9	
RL Laya (λ=1)	82.0 [76.7, 87.3]	2.0 [1.8, 2.3]	43.9	
RL Laya (λ=2)	83.1 [77.8, 88.4]	2.0 [1.8, 2.3]	43.9	

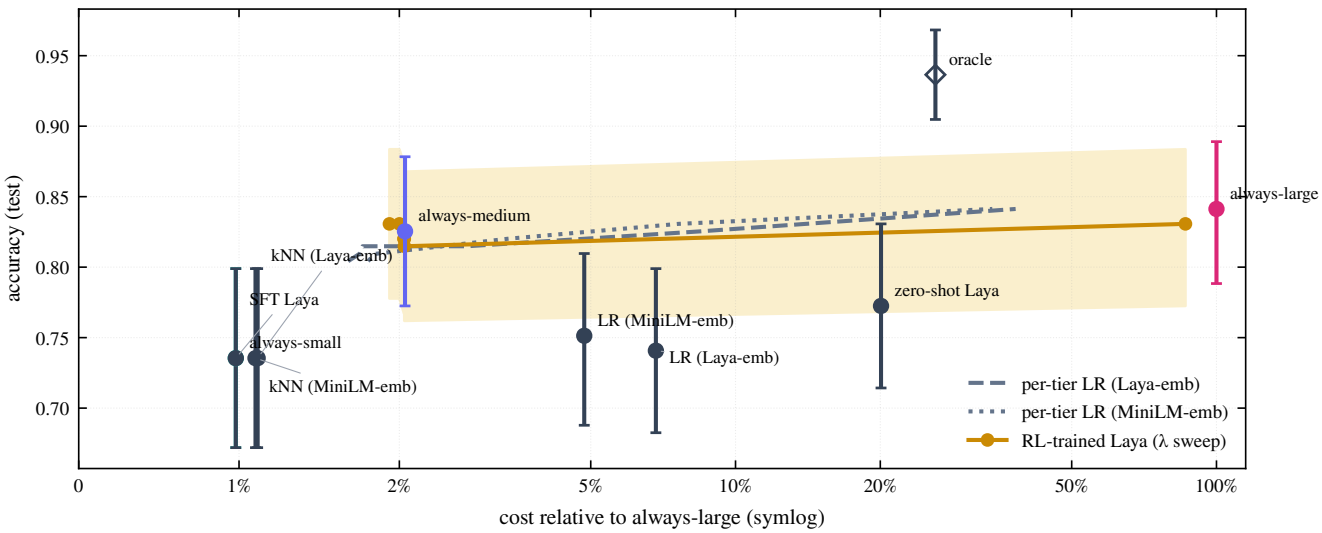


Figure 6. Accuracy vs. cost relative to always-large on the test split (symmetric-log cost axis). Shaded: 95% CI of the RL family; error bars: 95% CI; hollow diamond: hindsight oracle.

drives, with model `pennywise/auto`; Alan Code ran with Auto Router in the Alan desktop app. Every folder was re-scored by the same script with the same tests.

Pennywise merged 10 of 10 tasks, matching Auto Router's 10, and finished the set in 11.0 minutes versus 36.9, 3.4× faster (Table 6, Figure 13). Auto Router sent every task to the same model (DeepSeek V4.1 Flash). Without the cascade, Pennywise's `auto` policy merged only 7/10 (night) and 6/10 (daytime repeat): its cheap tier frequently broke OpenCode's tool loop with malformed tool

Table 4. RL policies vs. always-large on the test split: paired accuracy difference and relative cost with 95% bootstrap intervals, test action share (small/medium/large), mean reward, and the selected checkpoint.

λ	Acc %	Δ vs large (pts) [95% CI]	Cost % [95% CI]	S/M/L %	Reward	Best step
0	83.1	−1.1 [−3.7, +1.6]	86.2 [78.8, 92.4]	7/8/85	0.831	800
0.1	83.1	−1.1 [−6.9, +4.8]	1.9 [1.7, 2.2]	6/94/0	0.829	800
0.25	82.0	−2.1 [−7.4, +3.2]	2.0 [1.8, 2.3]	1/99/0	0.815	800
0.5	81.5	−2.6 [−7.9, +2.6]	2.0 [1.8, 2.3]	3/97/0	0.804	760
1	82.0	−2.1 [−7.4, +3.2]	2.0 [1.8, 2.3]	2/98/0	0.798	640
2	83.1	−1.1 [−6.9, +4.8]	2.0 [1.8, 2.3]	6/94/0	0.788	720

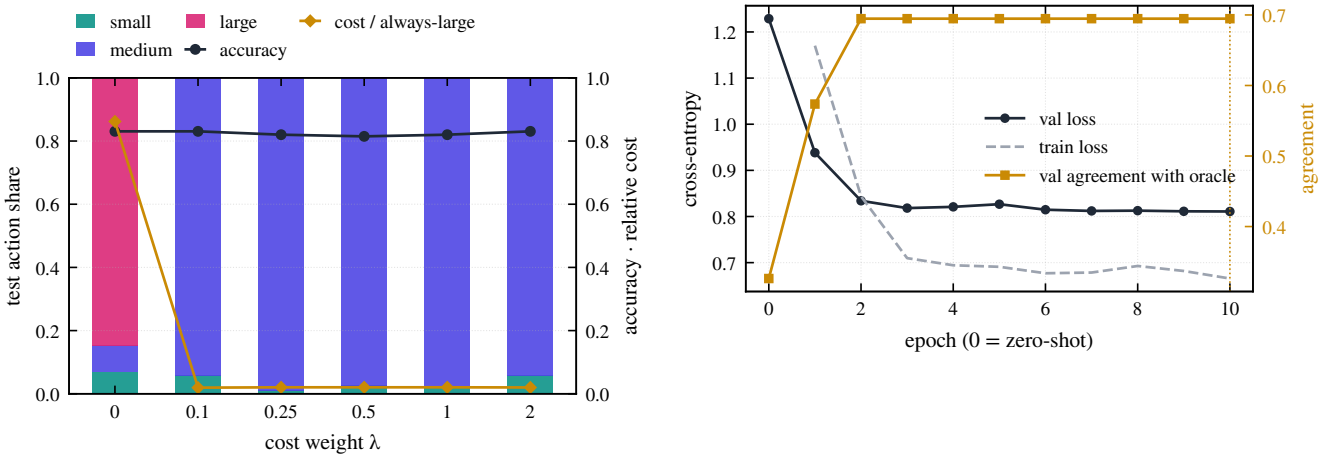


Figure 7. Left: effect of the cost weight, test action share (bars) with accuracy and relative cost (lines) for each trained λ. Right: supervised warm start, loss (left axis) and validation agreement with the oracle label (right axis), which rises from 32.6% zero-shot to 69.5% (epoch 10 dotted).

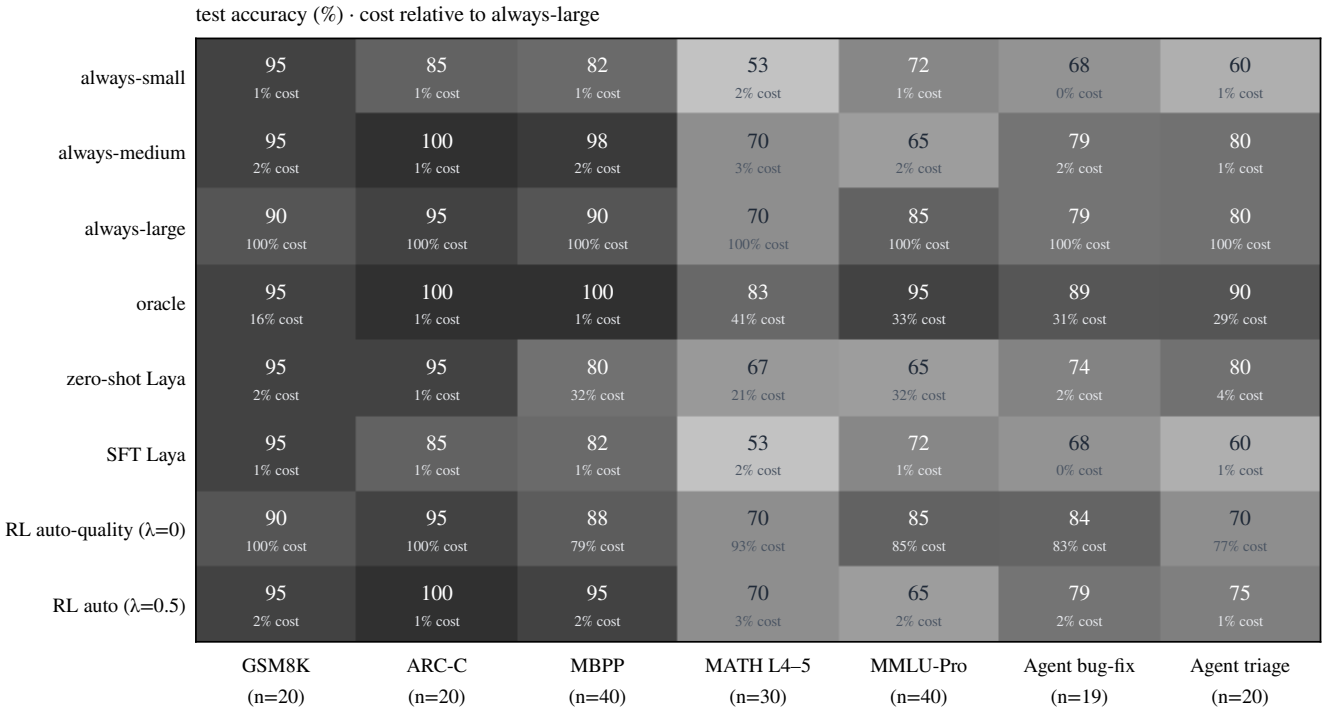


Figure 8. Per-source test accuracy (large digits, %) and cost relative to always-large (small text) for trivial policies, the oracle, Laya without RL, and the three RL modes. Agent-style slices are the two rightmost columns.

names, invalid arguments, empty replies and invented file paths. The cascade turns those failures into escalations; in the measured run 8 of 10 sessions escalated (45 cheap-tier and 81 top-tier calls in total), while easy sessions that ran cleanly stayed on the cheap tier (Figure 14).

9 System

The router adds one encoder pass, 43.9 ms median on the build machine's GPU for `auto`. The proxy is OpenAI-compatible: it routes once per conversation (keyed by the system prompt and first user message, or an `x-session-id` header) and keeps that tier for the session so agent tool loops are never re-routed mid-task; tools, tool choice and

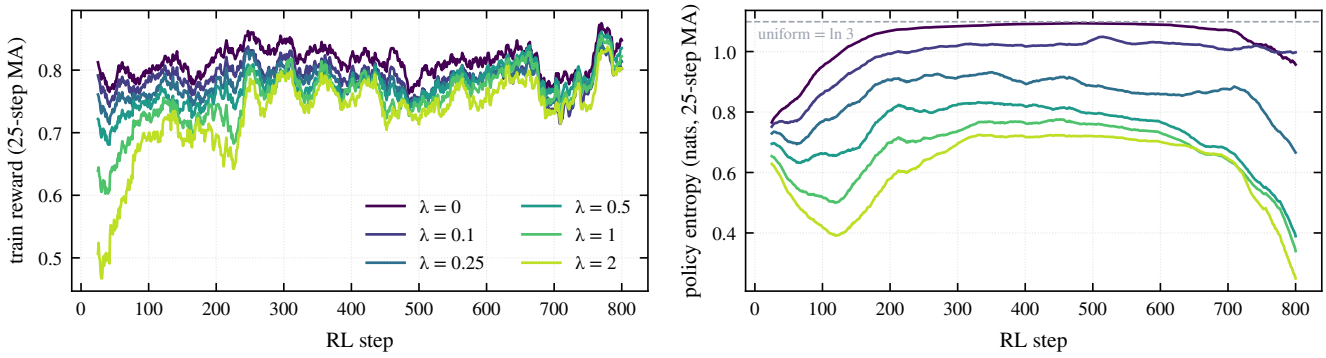


Figure 9. RL training dynamics per λ (25-step moving averages over sampled actions): reward (left) and policy entropy (right; dashed: uniform policy).

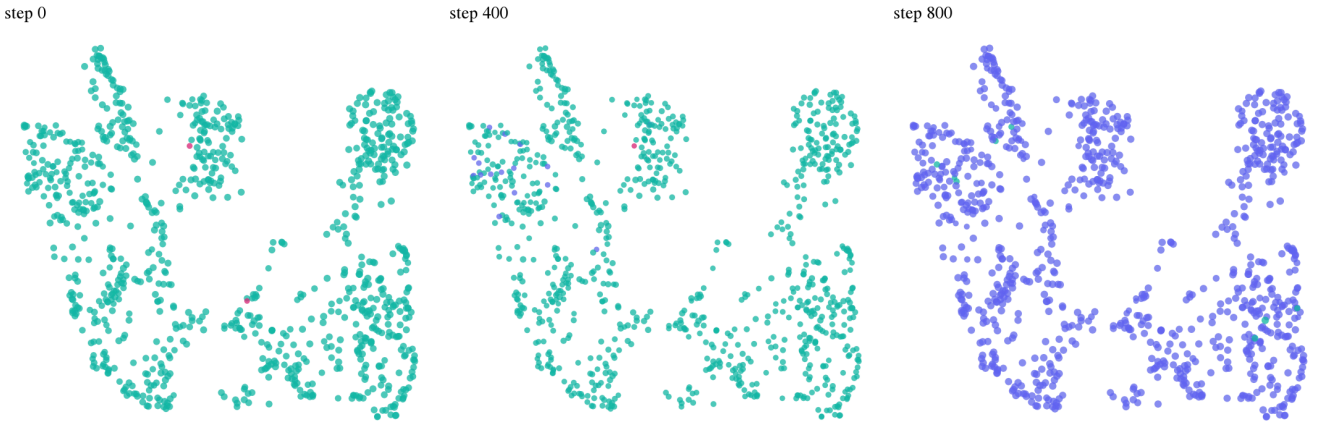


Figure 12. UMAP [17] of base-model prompt embeddings, coloured by the `auto` policy's chosen tier (teal = small, indigo = medium, pink = large; size = confidence) at the start, middle and end of RL.

tool messages pass through unchanged, including when streaming (Figure 15).

Agent before/after demo. We ran 4 small coding tasks (a bug or missing function plus a unit-test suite) with OpenCode, the open coding-agent CLI that Alan drives, once with the unrouted large tier and once with `auto`. "Merged" means the tests pass after the agent's edit. Cost per merged PR fell from \$0.0307 to \$0.0011 (27× lower at list price), while total wall-clock time rose from 1.9 to 65.6 minutes because the free medium-tier endpoint was slow that night (often over a minute per step). This is a small live demo, not a benchmark, and ran in the CLI directly rather than inside an Alan session (Table 7).

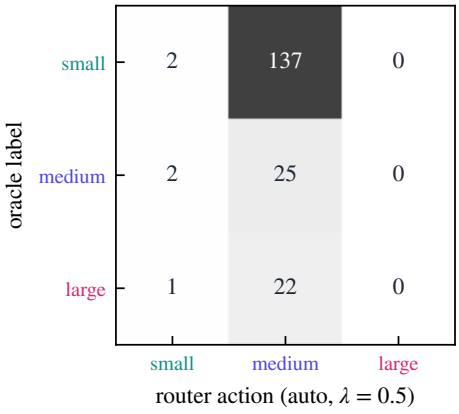


Figure 10. Router action vs. oracle label on the test split (counts) for `auto`.

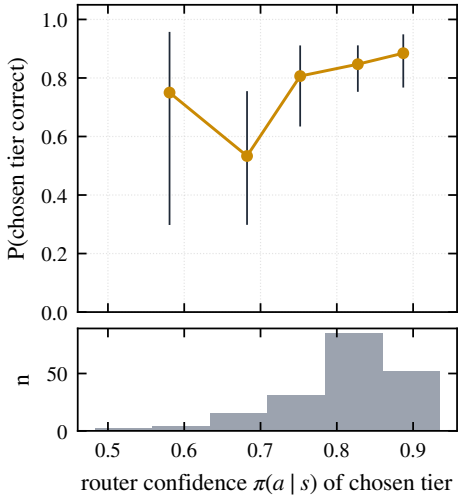


Figure 11. Reliability of the `auto` policy's confidence (bins with ≥ 3 prompts; Wilson 95% intervals) and its histogram.

10 Limitations and future work

The test set is small (n = 189), so intervals are wide; we use a single seed. The setting is offline: although RL uses only the sampled arm, the outcome table was collected with full information, which an online system would not have. Benchmark prompts are not production traffic, and the agent slice consists of synthetic single-function tasks. Costs are list prices of a free-API run, one of them a proxy price. Laya is known to degrade with many options [1], which is irrelevant at three tiers but limits extension to many models. Jev was not benchmarked. The agent benchmark has 10 tasks and one run per system;

Table 5. Per-source accuracy % / cost % of always-large on the test split.

Source	always-small	always-medium	always-large	RL Laya (λ=0.5)
Agent bug-fix (n=19)	68.4 / 0.3	78.9 / 1.7	78.9 / 100.0	78.9 / 1.7
Agent CI triage (n=20)	60.0 / 0.6	80.0 / 0.9	80.0 / 100.0	75.0 / 0.9
ARC-Challenge (n=20)	85.0 / 0.7	100.0 / 1.1	95.0 / 100.0	100.0 / 1.1
GSM8K (n=20)	95.0 / 1.1	95.0 / 2.2	90.0 / 100.0	95.0 / 2.2
MATH L4-5 (n=30)	53.3 / 1.7	70.0 / 3.0	70.0 / 100.0	70.0 / 3.0
MBPP (n=40)	82.5 / 0.7	97.5 / 2.1	90.0 / 100.0	95.0 / 2.1
MMLU-Pro (n=40)	72.5 / 1.0	65.0 / 1.9	85.0 / 100.0	65.0 / 1.9

Table 6. Agent coding benchmark (10 tasks, one run each). Merged = tests pass and the test file is unchanged. Top: the two systems compared; below: Pennywise configurations on the way there.

System	Merged	Easy	Medium	Hard	Total min
Alan Auto Router	10/10	3/3	3/3	4/4	36.9
Pennywise + cascade v0.2b	10/10	3/3	3/3	4/4	11.0
Pennywise large only	10/10	3/3	3/3	4/4	12.9
Pennywise + cascade→MiniMax v0.3	10/10	3/3	3/3	4/4	30.1
Pennywise + cascade v0.2	5/10	2/3	2/3	1/4	7.7
Pennywise auto (night)	7/10	3/3	2/3	2/4	66.1
Pennywise auto (day)	6/10	2/3	2/3	2/4	8.9

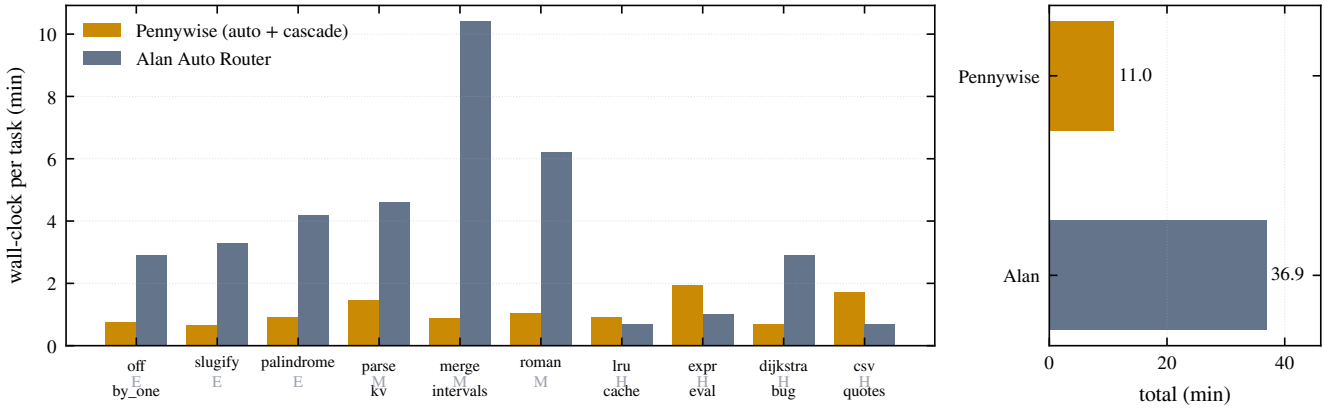


Figure 13. Wall-clock per task (left; letters mark easy, medium, hard) and in total (right) for Pennywise (router + tool-call cascade) and Alan Code's Auto Router. Both merged all 10 tasks.

the cascade's checks were designed after inspecting failures on these same tasks, so its result is optimistic until it is repeated on fresh tasks, while Auto Router received no tuning. We do not compare agent cost: Alan bills with prompt caching, which

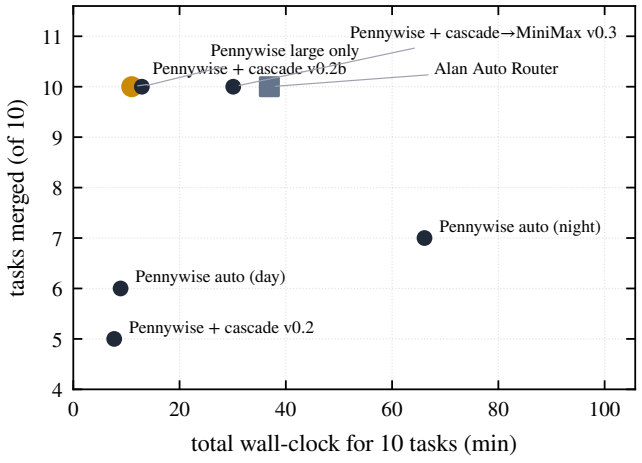


Figure 14. Tasks merged vs. total time for each configuration. The cascade moves Pennywise from 6–7 merged to 10 while keeping it fast.

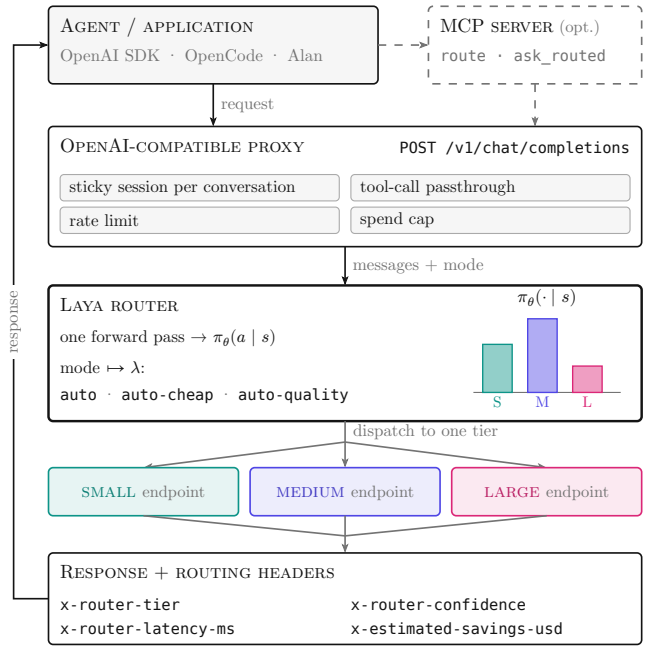


Figure 15. Serving path. Agents point an OpenAI SDK at the proxy; Laya routes once per conversation and the chosen tier answers; routing metadata is returned in response headers. An MCP server exposes the same router as tools; it is published on npm as `pennywise-mcp` (Node + ONNX Runtime, identical decisions to the Python router on our 31-prompt parity set) and runs inside the Alan desktop app as a custom connector.

Table 7. Agent before/after on 4 coding tasks (OpenCode CLI; list prices; free shared endpoints).

Model	Pass	Cost (USD)	USD / merged PR	Wall (min)
`pennywise/large`	4/4	0.1227	0.0307	1.9
`pennywise/auto`	4/4	0.0045	0.0011	65.6

> the free endpoints we used do not report, so the two cost figures are not on the same basis.

Planned work: partial-feedback online RL with a feedback endpoint; comparison with an exact expected-reward objective (possible offline because all outcomes are known); evaluation on RouterBench [6]; out-of-distribution tests; more tiers; multiple seeds; calibration.

11 Reproducibility

Code, configs and this report's sources are at commit `eb720a3-dirty`. `config/models.yaml` (tiers, prices) and `config/train.yaml` (splits, hyperparameters, λ grid, seed) define the run. Pipeline: `data/build_prompts.py` → `data/run_models.py` → `eval.features` → `eval.baselines` → `router.sft` → `router.rl` → `eval.evaluate` → `eval.export_viz` → `paper/build.py`. Every number, table and figure in this report is regenerated from the evaluation outputs by `paper/gen_tables.py` and `paper/gen_extra.py`.

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